

Capstone Project Proposal – Group 5

Workforce Burnout Prediction, Sick-Day Forecasting & HR Intervention Recommendation System

1. Introduction

Employee burnout has become a major challenge in modern organizations due to increasing workloads, high stress levels, and poor work-life balance. Burnout negatively affects employee wellbeing, productivity, absenteeism, and organizational performance. Many organizations struggle to identify burnout early enough to take preventive action.

This project proposes the development of a Machine Learning–Driven HR Analytics System capable of predicting employee burnout risk, forecasting sick days using regression techniques, and recommending appropriate HR interventions using a rule-based recommendation engine.

2. Problem Statement

Organizations often rely on reactive HR strategies after employee burnout has already affected performance and productivity. This results in increased absenteeism, reduced morale, higher turnover, and financial losses. There is a need for an intelligent system that can proactively identify burnout risks, estimate likely absenteeism, and recommend interventions before the situation worsens.

3. Project Objectives

Main Objective:

To develop a predictive HR analytics system for burnout prediction, sick-day forecasting, and HR intervention recommendations.

Specific Objectives:

1. Analyze HR employee data to identify burnout-related factors.
2. Develop a machine learning classification model for burnout prediction.
3. Develop a regression model to forecast employee sick days.
4. Design a rule-based intervention recommendation engine.
5. Build an interactive Streamlit dashboard for visualization and prediction.
6. Apply explainable AI techniques to improve model transparency.

4. Scope of the Project

The project will focus on HR analytics using employee-related attributes such as Job Satisfaction, Work-Life Balance, Overtime, Monthly Income, Years at Company, Performance Rating, and Environment Satisfaction.

The system will include the following core modules:

- ❖ Burnout Risk Prediction (Classification)
- ❖ Sick-Day Forecasting (Regression)
- ❖ HR Intervention Recommendation Engine (Rule-Based System)
- ❖ Interactive Dashboard and Data Visualization

5. Data Understanding

The project will use the IBM HR Analytics Employee Attrition & Performance Dataset from Kaggle. The dataset contains approximately 1,470 employee records and 35 attributes.

Key features include:

- JobSatisfaction
- WorkLifeBalance
- Overtime
- MonthlyIncome
- YearsAtCompany
- EnvironmentSatisfaction
- PerformanceRating
- JobRole

6. Methodology

6.1 Data Preparation

1. Handling missing values
2. Encoding categorical variables
3. Feature scaling and normalization
4. Feature engineering
5. Train-test splitting

6.2 Exploratory Data Analysis (EDA)

EDA will be conducted using visualizations such as correlation heatmaps, burnout distribution charts, feature importance plots, and regression analysis graphs.

7. System Design

Stage 1: Burnout Prediction (Classification)

The first stage of the system will classify employees into Low, Medium, or High burnout risk levels using supervised machine learning classification algorithms.

Planned models include:

- Logistic Regression(baseline)
- Random Forest Classifier
- XGBoost Classifier

Stage 2: Sick-Day Prediction (Regression)

The second stage will estimate the number of sick days an employee is likely to take due to burnout and workplace stress factors. Since the dataset does not contain sick-day records, a synthetic target variable will be engineered using burnout-related features.

Regression models include:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

Stage 3: HR Intervention Recommendation System

The third stage will use a rule-based recommendation engine to provide personalized HR interventions based on employee burnout risk and workplace conditions.

Example recommendations include:

- Reduce overtime workload
- Flexible scheduling
- Employee wellness programs
- Counseling and mental health support
- Mandatory leave recommendations

8. Model Evaluation

The classification models will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

The regression models will be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

9. Deployment

The final system will be deployed as a Streamlit web application. Users will be able to upload employee datasets, view burnout predictions, forecast sick days, and receive HR intervention recommendations.

10. Tools and Technologies

- Python
- pandas
- NumPy
- scikit-learn
- XGBoost
- SHAP
- Matplotlib
- Seaborn
- Streamlit
- GitHub
- Jupyter Notebook
- VS Code

11. Project Timeline (3 Weeks)

Week	Activities
Week 1	Data collection, preprocessing, feature engineering, and exploratory data analysis. Development of baseline burnout prediction model.
Week 2	Advanced model development for burnout classification and sick-day regression. Hyperparameter tuning and SHAP explainability.
Week 3	Development of rule-based recommendation system, Streamlit dashboard deployment, testing, documentation, and final presentation.

12. Expected Deliverables

- ✓ Cleaned HR analytics dataset
- ✓ Burnout prediction classification model
- ✓ Sick-day forecasting regression model
- ✓ Rule-based intervention recommendation system
- ✓ Interactive Streamlit dashboard
- ✓ Final project report and presentation

13. Conclusion

This project demonstrates how machine learning and HR analytics can be combined to proactively address employee burnout. By integrating burnout prediction, sick-day forecasting, and intervention recommendations into one intelligent platform, the system aims to support data-driven HR decision-making and improve employee wellbeing.